

FLIGHTRAX

Project Application

FlightraX is WCC's web-based system for automated flight scheduling, digital permission routing, public flight tracking monitors, and role-based approvals for students and flight instructors.

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System documentation. Describes what the application is, who it serves, what it offers, and how its business lifecycle works. This is not a user manual.

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FlightraX Team (Researchers)	Web + installable app (PWA)	Feature-complete MVP, in QA

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1 Introduction

1.1 What FlightraX is

FlightraX is a web-based flight operations and training management system built for WCC's flight school. It digitises the paperwork and coordination that surround every training flight: the flight plan, the weight and balance sheet, the instructor's approval, the daily schedule board, and the live status of the fleet.

The system is delivered as a responsive web application that can also be installed on a phone or tablet as a standalone app. It runs in the browser on desktop and mobile, sends push notifications to installed devices, and drives a public flight monitor for a lobby television that needs no login.

Everything in FlightraX is organised around a small set of people with clearly separated responsibilities: students who file and fly, instructors who review and supervise, three administrative departments that keep the fleet, the accounts, the airspace notices, and the schedule in order, and a single superadmin who owns the administrative accounts and the system's configuration.

1.2 The problem it replaces

Before FlightraX, the school's flight day ran on paper and on people remembering to tell each other things:

- **Flight plans** were handwritten on the CAAP Form ATS 2019-1 and walked to an instructor for a signature. A missing signature, an expired license, or a mis-typed departure time was only caught by the person reading the form.
- **Weight and balance** was computed by hand from the aircraft's figures, and the sheet lived only on the clipboard it was written on.
- **Permission to fly** was a verbal or signed handshake with no record of who approved what and when, and no way for the tower desk to know a flight was cleared.
- **The daily schedule** was an Excel sheet or a whiteboard. Students found out it was ready by asking or walking past it.
- **Fleet status** (who is airborne, who has landed, which aircraft is grounded) lived in the heads of whoever was on the ramp.
- **Accounts and credentials** were verified by looking at a physical ID, with nothing tying a person's license to the flights they were allowed to command.

FlightraX replaces each of these with a single, role-aware system of record: filed forms are validated before they can be submitted, approvals are signed digitally and audited, the schedule is posted once and everyone is told, and the fleet's status is visible to everyone who needs it, including the public monitor in the lobby.

1.3 Product pillars

Schedule

A posted daily board of the fleet, by aircraft and half-hour, with the school's own Excel workbooks uploaded alongside it. One ping tells everyone it is ready.

Route

Digital permission routing. A flight request moves from draft to pending approval to approved or rejected, through the right instructor or pilot-in-command lane, with signatures and licenses captured at each step.

Track

Live flight status boards inside the app and a public flight tracking monitor for the lobby TV: on ground, departed, arrived, with NOTAMs underneath.

Approve

Role-based approvals for accounts and flights, protected by a personal four-digit passcode and backed by a full notification trail.

2 Goals and Objectives

The project has one overarching goal: make WCC's training flight day run on a shared, trustworthy record instead of paper and memory, without adding cost or complexity for a small school.

2.1 Primary goals

Goal	What it means in FlightraX
Automated flight scheduling	A schedule board whose rows are the fleet itself, posted by Flight Operations Personnel, readable by everyone, and announced with a single notification. Uploaded Excel workbooks appear on a month calendar and open in a built-in viewer.
Digital permission routing	Every flight is a request that carries a complete CAAP flight plan and a weight and balance sheet. It is routed to the assigned flight instructor and to the pilot in command, either of whom may approve or reject it with a signed, passcode-confirmed decision.
Public flight tracking monitors	A no-login monitor page for the lobby television that rotates through On Ground, Departed, and Arrived screens, refreshed every ten seconds, with active NOTAMs on a band beneath.
Role-based approvals	Student and instructor accounts are verified by Flight Operations Personnel before they can use the system. Administrative accounts are issued by the superadmin. Flight requests are approved by instructors. NOTAMs are posted by Safety Personnel. Each role sees only the pages and actions that belong to it.

2.2 Supporting objectives

- **Correct by construction.** Forms reject invalid ICAO field formats, missing signatures, expired licenses, grounded aircraft, double-booked aircraft, and unavailable instructors before anything reaches a reviewer.
- **Auditable.** Filed forms snapshot the aircraft, pilot, licenses, and signatures at the moment of filing and approval, so later edits to profiles or the fleet never rewrite history. Every approval, rejection, commencement, termination, and cancellation records who did it and when.
- **Safe on a slow connection.** Pressing a button twice, or retrying after the connection drops, never creates a second record, never notifies people twice, and never shows an error for something that already succeeded.
- **Plain language.** Every error a user sees is a sentence they can act on. Database and system messages never reach the screen.
- **Mobile first.** Every surface works on a phone. The app installs to the home screen and receives push notifications, so a student learns their request was approved without opening anything.
- **Zero running cost.** The whole system fits inside the free tiers of its hosting providers, and its operating limits are documented so the school knows when and why an upgrade would be needed.
- **Official paperwork on demand.** Any approved flight can be downloaded as a filled CAAP Form ATS 2019-1 and a weight and balance sheet, with signatures and license lines printed on them.

2.3 Success criteria

- A student can go from registration to an approved, flown, and logged flight without touching paper.
- An instructor sees a new request the moment it is submitted and can decide it from a phone.
- Flight Operations Personnel can verify accounts, maintain the fleet, mark instructor leave, and post the day's schedule from one place.
- The tower desk (Air Traffic Controller department) can see every approved flight and its live status without asking anyone.
- The superadmin can stand up, hand over, or recover every administrative account without a developer touching the database.
- The lobby monitor shows the fleet's status to anyone walking past.

3 Scope and the MVP

The minimum viable product is the complete flight day for a small school: accounts, fleet, flight documents, approvals, live operations, schedule, notices, and notifications. All of it is built. The items below are what the MVP covers and, separately, what was deliberately left out.

3.1 In scope (delivered)

Area	Delivered capability
Accounts	Email and password sign-in per role. Student and instructor registration with ID number and ID image, reviewed by Flight Operations Personnel. Administrative accounts are not self-registered: the superadmin creates each admin with a department and a one-time temporary password, and the account must set its own password on first sign-in. Forgot and reset password by cross-device email link. Pending and rejected account states with resubmission.
Administration	A superadmin role, provisioned by a seeding command rather than a registration form. It manages the Admins directory: create admin accounts, hand over credentials with a copyable message, regenerate a temporary password as a recovery path, and see which admins still use one. A global navigation setting confines or restores the superadmin's own reach.
My Account	Profile photo, drawn signature, four-digit passcode, licenses with front and back images and ratings, certificates with image, personal flight log, change password.
Fleet	Aircraft types with ICAO designator and weight and balance specifications (arms, MTOW, baggage areas). Aircraft with registration mark and number, serial, colour and markings, photo, and status (active, maintenance, grounded, retired). Per-aircraft basic empty weight, arm, and moment.
Flight documents	CAAP Form ATS 2019-1 flight plan with full ICAO field validation, a dedicated date of flight, and auto-filled Other Information. Weight and balance sheet with server-computed totals and CG. Draft, edit, delete, submit, withdraw. PDF export onto the official form.
Flight requests	Review queue for instructors (assigned and all). Approve or reject with passcode, signature, and license checks. Self-approval for an eligible pilot in command. Rejection with reason and resubmission.
Flight operations	Dashboard flight status board and today's flights drawer scoped to each flight's participants. Commence, terminate, and cancel with passcode. Automatic standby after arrival and automatic no-show cancellation. Delayed and past-EET alerts.
Flight plans audit	Air Traffic Controller view of every completed or cancelled flight with search, status filter, and document download.
Schedule	Day board per aircraft with eleven session types, people pickers, overlap protection, and a "ping everyone" announcement. Files tab with Excel workbook upload, month calendar, and in-app sheet viewer.

People	Students and instructors directories with license and certificate details for staff, and a superadmin-only Admins directory. Instructor unavailability periods managed by Flight Operations Personnel.
NOTAMs	Posted by Safety Personnel with severity and expiry. Shown on dashboards and the public monitor. Active and expired tabs.
Notifications	Seventeen event types delivered in-app (bell, panel, page) in real time and as push notifications to installed devices, with a strict recipient matrix.
Public monitor	Lobby TV board at /monitor , no login, rotating screens, NOTAM band, Philippine time by default with a zulu toggle, fullscreen mode, and a sample-data preview.
Safe submission	Every form and action is protected against double taps and retries on a slow connection, on the device and on the server.
Installable app	Web app manifest, service worker, install prompt, standalone launch, badge icon.

3.2 Out of scope and deferred

Item	Decision
License and certificate expiry notifications	Dropped. The credential surfaces already show expiry state; a notification would restate what is on screen and cost a scheduled job.
Per-entry schedule notifications	Dropped by design. The board is announced once with a deliberate ping, like pinning up the paper board.
Notifying all instructors about every request	Rejected. Only the assigned instructor and the pilot in command are told.
Fleet-wide flight status notifications	Rejected. Per-flight events reach only that flight's participants; fleet awareness is what the monitor is for.
Actions on the dashboard status board	The board is monitoring only. Commence, terminate, and cancel live in the today's flights drawer.
Admin and superadmin self-registration	Removed in this revision. Administrative accounts exist only through the superadmin (admins) and the seeding command (the superadmin itself), so there is no public path into a privileged role.
iOS push verification	Written to iOS 16.4+ behaviour but untested on a device as of this document. Android and desktop are verified.
Billing, payments, student progress records, maintenance logs	Not part of the product.
User manual and training guide	Planned as a separate document after this one.

4 Users and Roles

4.1 Role catalogue

FlightraX has four application roles. The admin role is further divided into three departments, each with its own remit. Department is fixed when the account is created.

Role	How the account exists	Who they are and what they are for
Student	Registers; reviewed by Flight Operations Personnel	A trainee pilot. Files flight plans and weight and balance sheets, submits requests to an instructor, flies approved flights, and keeps licenses, certificates, signature, and passcode on their account.
Instructor	Registers; reviewed by Flight Operations Personnel	A flight instructor. Everything a student can do, plus reviewing flight requests they are assigned to, acting as pilot in command, and managing the live flights they are part of.
Admin: Flight Operations Personnel	Created by the superadmin; usable immediately	Runs the school's operations desk. Reviews accounts, manages aircraft types and aircraft, sets instructor unavailability, posts and pings the schedule, uploads schedule files.
Admin: Air Traffic Controller	Created by the superadmin; usable immediately	The tower desk. Sees the organised flight board, is notified of every approved flight, and audits completed and cancelled flight plans with their documents.
Admin: Safety Personnel	Created by the superadmin; usable immediately	Posts and removes NOTAMs (Notices to Airmen) that appear on dashboards and the public monitor.
Superadmin	Seeded by a command; exactly one account	The system owner's account. Creates and recovers admin accounts, holds every permission, and controls the global configuration, including how much of the application its own navigation exposes.

Entry points

- **Students and instructors** use the public role picker on the login and register pages.
- **Admins** are deliberately absent from the public picker. Their login page opens by pressing and holding the plane icon on the login page until it "takes off". Admins have no registration page: their accounts are issued by the superadmin.
- **The superadmin** has no visible entry at all. Its login page opens by tapping the FlightraX logo on the home page five times in quick succession, and there has never been a registration form for it; the account comes from the seeding command.
- Signing in on the wrong role's page is refused. A student cannot sign in through the instructor page even with valid credentials, and only the superadmin can sign in through the superadmin page.

4.2 Permission model

Access is expressed as named permissions attached to roles, with department permissions added on top for admins. Routes and navigation items are gated by the same permissions, so a role never sees a link it cannot open.

Permission	Student	Instructor	Flight Ops	ATC	Safety
account.view (My Account)	✓	✓	✓	✓	✓
dashboard.view (Home, Notifications)	✓	✓	✓	✓	✓
flight_documents.view	✓	✓	—	—	—
flight_requests.view (review queue)	—	✓	—	—	—
flight_plans.view (audit)	—	—	—	✓	—
instructors.view	✓	✓	✓	✓	✓
instructors.manage (availability)	—	—	✓	—	—
students.view	—	✓	✓	✓	✓
credentials.view_details	—	✓	✓	—	—
accounts.review	—	—	✓	—	—
aircrafts.view (fleet management)	—	—	✓	—	—
schedule.view	✓	✓	✓	✓	✓
schedule.manage (post, edit, ping, upload)	—	—	✓	—	—
notams.view (NOTAMs page)	—	—	—	—	✓
notams.manage (post, delete own)	—	—	—	—	✓
admins.view (Admins directory)	—	—	—	—	—

The superadmin is not a column because it passes every permission check by design. That is also how superadmin-only surfaces are expressed: admins.view is granted to no role at all, so only the superadmin ever satisfies it. Students, instructors, and Air Traffic Controllers still see NOTAMs on their dashboard home even though they hold no NOTAMs page permission. Flight Operations Personnel do not, because their home is the aircraft manager.

Holding every permission does not mean the superadmin's navigation shows everything. A global setting, **superadmin_full_navigation**, decides how much of the application the superadmin's own menu and routes expose; it ships turned off. Section 5.9 and Section 11.2 describe it.

4.3 Page access matrix

Page	Student	Instructor	Flight Ops	ATC	Safety
Home (/dashboard)	flight board	flight board	aircraft manager	organised board	organised board
My Account	✓	✓	✓	✓	✓
Flight Documents	✓	✓	—	—	—

Flight Requests (queue)	—	✓	—	—	—
Flight Plans (audit)	—	—	—	✓	—
Schedule	read	read	manage + ping + files	read	read
Instructors	list only	with credentials	credentials + availability	list	list
Students	—	✓	✓	✓	✓
Admins	—	—	—	—	—
Account Review	—	—	✓	—	—
NOTAMs page	—	—	—	—	✓
Notifications	✓	✓	✓	✓	✓
Public monitor (/monitor)	Public. No login required.				

The Admins page is superadmin-only, which is why no role holds it. The superadmin itself reaches every page when `superadmin_full_navigation` is on. When it is off (the shipped default), the superadmin sees and reaches only Home, Instructors, Students, Admins, My Account, and Notifications; typing any other URL redirects to the dashboard. A user who types a URL they cannot access is redirected to their dashboard. An account that is not yet approved can reach only the pending-approval page.

4.4 What each role can do

Student

- Register with student ID number and ID image; wait for approval; resubmit if rejected.
- Maintain My Account: photo, signature, passcode, licenses (with ratings and images), certificates, password.
- File a flight plan on an available aircraft, choose a flight instructor, choose a pilot in command (themselves with any valid license, Student Pilot License included), fill the weight and balance sheet, and submit the request.
- Edit or delete a draft or rejected plan; withdraw a pending request back to draft.
- Self-approve their own request when they are the pilot in command **and** hold a valid Private Pilot License. A student flying on a Student Pilot License can be the PIC, but their request must still be approved by the assigned instructor.
- Commence, terminate, and cancel their own flights from the today's flights drawer, with passcode.
- Read the schedule board and uploaded schedule files. See the instructors list and availability. Download their flight documents as PDF. Read their flight log and notifications.

Instructor

- Everything a student can do, with a wider approval rule: any active, non-expired license lets an instructor decide a request as pilot in command.
- Review flight requests: an "Assigned" queue (requests where they are the instructor or the PIC) and an "All" queue of every pending request. Approve or reject only those they are assigned to.
- Commence, terminate, or cancel the flights they are part of — as filer, pilot in command, or assigned instructor. Flights they have no part in are monitoring-only, like everyone else's.
- See the students directory with license and certificate details, and fellow instructors' credentials.

Admin: Flight Operations Personnel

- Home is the aircraft manager: create aircraft types and their weight and balance specifications; create, edit, photograph, re-status, and delete aircraft; set each aircraft's basic empty weight figures.
- Account Review: approve or reject student and instructor registrations, with a reason on rejection, and see queue metrics.
- Instructors: add and remove unavailability periods, which block that instructor from being chosen on flight plans for those dates.
- Schedule: post, edit, and delete entries per aircraft per day; ping everyone when the board is ready; upload and remove Excel schedule workbooks.

Admin: Air Traffic Controller

- Home is the organised flight board: today's flights grouped by status, with NOTAMs.
- Flight Plans audit: search and filter completed and cancelled flights, open the flight log with plan, weight and balance, and journey, and download the documents.
- Receives a notification for every approved flight request.

Admin: Safety Personnel

- Home is the organised flight board with NOTAMs.
- NOTAMs page: post a NOTAM with title, description, severity, and expiry date; delete NOTAMs they posted; browse active and expired notices.

Superadmin

- Exists as exactly one seeded account; there is no registration path. The seeding command creates it on a fresh database and, run again, resets it to a known state as a recovery path, after which its password must be changed on the next sign-in.
- Admins page: create admin accounts with a chosen department. Each new account is issued a neutral, randomly generated temporary password, shown once behind a reveal toggle, together with a copyable handoff message for the new admin.
- Regenerate a temporary password for any admin who is locked out; the old password stops working immediately and a fresh handoff message is produced.
- See at a glance which admins still sign in with a temporary password and which have set their own.
- Sees the students and instructors directories, its own account, and notifications. With `superadmin_full_navigation` turned on, every other page and route opens as well.
- Passes every permission check, so any future superadmin-only surface is expressed simply as a permission no role holds.

5 System Modules and Features

The application is organised into twelve feature modules. Each owns its own screens, validation, and server logic, and none imports from another. This section describes what each module offers to its users.

5.1 Authentication and My Account

Sign-in and registration. Email and password authentication with a separate login page per role. Registration exists only for students and instructors: it collects a full name in "Lastname, First M." format, email, password, an ID number, and an ID image (JPG, PNG, or WebP, 5 MB maximum, automatically shrunk in the browser when larger than 1 MB). An ID number that is already registered is refused before any account is created. Registration ends on the pending-approval page. Administrative roles have no registration form at all: admin accounts are created by the superadmin (Section 5.9), and the superadmin account by the seeding command (Section 6.1).

Temporary passwords and the forced change. An account created or reset with a known password — a new admin, a regenerated admin password, or a re-seeded superadmin — carries a must-change flag. On sign-in, a dialog that cannot be dismissed, skipped, or clicked away asks for a new password (and only the new password, since the temporary one was just typed to get in). Setting it clears the flag and unlocks the session.

Password recovery. A forgot-password form sends a reset link that carries a token hash rather than a browser-bound code, so the link works when opened on a different device from the one that requested it. The response wording is identical whether or not the email exists. Admins locked out of a temporary password are instead recovered by the superadmin regenerating it.

Pending and rejected states. An unapproved account can reach only the pending-approval page. A rejected account sees the reviewer's reason and a resubmission form (corrected name, ID number, optional replacement ID image). Both states offer push notification opt-in so the person learns the outcome on their phone.

My Account holds everything a person needs before they can fly:

- **Profile photo**, shown in the sidebar, header, and directories.
- **Signature**, drawn on a canvas and stored as vector SVG. It is copied onto every flight plan and weight and balance sheet the person files or approves.
- **Passcode**, a four-digit security code (students and instructors) required to approve, reject, commence, terminate, or cancel a flight.
- **Licenses**: Student, Private, Commercial, or Flight Instructor license with number, ratings (Instrument Rating plus a type rating for every aircraft type in the fleet), expiry date or no-expiry flag, status, and front and back images. Validity drives what the person may file and approve. The same license type and number cannot be added twice.
- **Certificates**: free-form credentials with title, description, optional expiry, and an image.
- **Flight log**: the person's own completed and cancelled flights.
- **Change password**.

A student or instructor with no license on file is prompted with a "No Licenses Detected" dialog on the dashboard until one is added.

5.2 Account Review

The queue Flight Operations Personnel use to verify new students and instructors. Each request shows the person's name, email, ID number, and a preview of the uploaded ID image, plus counts of pending, approved, and rejected requests per role. Approval opens the account; rejection requires a reason, which the person sees on their pending page. A request already decided by a colleague cannot be decided twice. Resubmitted requests return to the queue and the reviewers are notified.

5.3 Fleet (Aircraft Types and Aircraft)

Aircraft types are the fleet's catalogue: a name (Cessna 152, Tecnam P2002JF), an ICAO designator printed on flight plans, and the weight and balance specifications shared by every aircraft of that type: usable fuel arm, instructor and student seat arm, maximum takeoff weight, baggage area maximum weight, and up to six baggage areas with their arms. A type in use cannot be deleted.

Aircraft carry a registration mark (RP-C1884) and number, type, serial number, colour and markings, an optional photo, remarks, and a status: Active, Maintenance, Grounded, Retired. Each aircraft also has its own basic empty weight, arm, and moment. Registration marks are unique.

An aircraft becomes selectable for flight plans only when it is active, its type has complete specifications, and its own basic empty weight is set. Until then the picker labels it "Needs admin setup". A status change away from Active notifies the participants of every live flight scheduled on that aircraft. Fleet management is the home page of Flight Operations Personnel.

5.4 Flight Documents

The heart of the system. A flight document set is one flight plan plus one weight and balance sheet, bundled into a flight request.

Flight plan follows CAAP Form ATS 2019-1 (the ICAO flight plan). The form covers addressee and originator; the date and time of filing as DDHHMM; the **date of flight**, its own required field in ICAO YYMMDD form, which prints as the DOF/ line in Other Information and is the day every availability check, conflict check, and board keys on; flight rules; type of flight; number of aircraft; wake turbulence category; COM/NAV and surveillance equipment; departure and destination aerodromes chosen from a Philippine aerodrome list (or ZZZZ for the school's home field, RPT-20 Binalonan); departure time in zulu, pre-filled to one hour after the filing time and freely editable; cruising speed and level; route; total estimated elapsed time; alternates; Other Information; endurance; persons on board; emergency radio, survival equipment, jackets, and dinghies; aircraft colour and markings; remarks; pilot in command; and the assigned flight instructor. Every field is validated against its ICAO format with a plain-language message. The Other Information block is auto-filled with DOF, DEP, DEST, ALTN, and license lines and must include the departure and destination locations.

The emergency and survival equipment block works the way the paper form does: the filer crosses out what is *not* carried, and whatever is left unmarked is declared as carried — the same muscle memory as striking through boxes on the printed CAAP form. Endurance may be left blank while the plan is a draft; submission refuses until it is filled, with a message saying exactly that.

Filing snapshots the aircraft's identification, type, designator, and markings; the filer's name, signature, and licenses; and generates a unique plan code in the form **FP-YYMMDD-NNNN**.

Weight and balance is filled after the plan. The filer enters usable fuel weight and moment, instructor-plus-student weight and moment, and a weight and moment for each baggage area of the type. The server adds the aircraft's basic empty weight figures, computes total weight, total moment, and centre of gravity, and grades the result Within Limits or Overweight against the type's maximum takeoff weight. The filer records the

balance status (Balanced, Nose Heavy, Tail Heavy). The sheet is signed by the preparer at filing and by the approver at approval.

Lists and tabs. A person's own documents are grouped as In Progress (draft and rejected), Pending, and Approved, searchable by plan code. Each card shows the aircraft, route, date of flight, status, and whether the sheet exists.

Documents download. Any flight can be exported as the flight plan overlaid on the official CAAP form PDF, the weight and balance sheet, or both in one file. No signature prints unless the flight is approved: once it is, the Pilot's Name and Signature column carries the pilot in command — name, drawn signature, and license line (for example "123456-CPL | C152/IR | 15 MAR '25") from the snapshot taken at filing. Draft, pending, and rejected plans print an empty signature block under a DRAFT watermark, and the app says so wherever they are previewed. The Duly Authorised Representative column is never printed; the approval itself is recorded in the database, not on the form. Generation happens in the browser.

5.5 Flight Requests (review)

The instructor's queue. "Assigned" lists pending requests where the viewer is the flight instructor or the pilot in command; "All" lists every pending request for awareness. The queue updates in real time when a student submits. A review page shows the plan, the weight and balance sheet, and the journey, and offers Approve and Reject. Both require the reviewer's passcode; approval also requires a stored signature and a valid license, and copies both onto the documents. Rejection requires a written reason. A requester who is also an eligible reviewer sees a Self-approve action.

5.6 Flight Operations (Dashboard)

Flight status board. The home surface for students and instructors. One row per active aircraft showing its current journey: on ground (scheduled today), active (departed), or arrived, with trainee, PIC, instructor, route, departure time, and elapsed time. A scheduled flight whose date of flight has passed is flagged Delayed; an active flight airborne longer than its filed EET is flagged. The board is monitoring only and updates in real time. A NOTAM section and a live zulu clock sit alongside it.

Organised board. The home surface for Air Traffic Controller and Safety Personnel: the same data grouped by status.

Today's flights drawer. Lists the flights the viewer is part of today — as filer, pilot in command, or assigned instructor; the superadmin sees them all — and carries the three operational actions, each confirmed with the passcode: **Commence** (on ground to active), **Terminate** (active to arrived), and **Cancel** (on ground to cancelled). The same participant rule guards the actions on the server: only a flight's own people can move it.

5.7 Flight Plans Audit

The Air Traffic Controller's record of every flight that reached a terminal state: arrived, standby, or cancelled. Searchable by aircraft, trainee, or PIC; filterable by completed or cancelled; each entry opens the full flight log with documents download.

5.8 Schedule

Board view. A day board whose rows are every aircraft in the fleet, grouped by type, against a 24-hour half-hour time axis that opens at 06:00. Flight Operations Personnel click a cell to post an entry: start and end time, one of eleven session types (Pre-Solo Simulator, Taxi Exercise, Local Flight, Cross Country, Instrument Simulator, CAAP Checkride, Company Flight, Out of Service, To Be Determined, Cross Trainees, UPRT), a pilot

and/or an instructor chosen from approved people, or a free-text note for Out of Service and To Be Determined. Entries on one aircraft cannot overlap. A legend explains the colours. A day strip and month picker navigate dates, carried in the URL.

Ping everyone. When the day is ready, the admin sends one notification to every approved user. Re-pinging is allowed; the confirmation warns when a ping already went out and shows who sent it.

Files view. The school's existing Excel workbook can be uploaded as-is (.xlsx, 4 MB maximum, covering up to two months, no overlap with another file). The server parses every sheet into a grid with values, merges, fills, and alignment, so the viewer renders it without a spreadsheet library. A month calendar paints every covered date; opening a date opens the workbook at that sheet. The original file can be downloaded. Uploading notifies everyone.

5.9 People Directories (Instructors, Students, Admins)

Server-paginated, searchable tables of approved instructors and students, with license and certificate tags that open detail dialogs (including the credential images) for roles that hold the credentials permission. The instructors table also shows each instructor's upcoming unavailability. Flight Operations Personnel manage unavailability periods here: date ranges that cannot overlap, cannot be entirely in the past, and self-prune once they have passed. Adding a period notifies every student.

The **Admins directory** is the superadmin's counterpart, visible to no one else. It lists every admin with name, email, department, join date, and a password column that shows whether the account still uses its temporary password or has set its own. From here the superadmin creates a new admin — name, email, department — and receives the generated temporary password exactly once, obscured behind a reveal toggle and never editable, together with a copyable handoff message that names the login page. A per-row action regenerates the temporary password for a locked-out admin, invalidating the old one and producing a fresh handoff message. The table searches by name or email and paginates on the server like the other directories.

5.10 NOTAMs

Notices to Airmen posted by Safety Personnel: a title (200 characters), a description (300 characters, sized to stay readable on the lobby TV), a severity (Advisory, Warning, Alert), and an expiry date no earlier than today. A NOTAM is active until the end of its expiry date in Philippine time and then moves to the Expired tab. Active NOTAMs appear on dashboards, in real time, and on the public monitor's band. Posting notifies everyone except Flight Operations Personnel, who have no NOTAM surface. A NOTAM can be deleted only by its author.

5.11 Notifications

An in-app feed and a push channel fed by the same event rows. On desktop a bell in the header carries the unread count and opens a panel; on mobile it opens a full notifications page. Rows deep-link to the relevant page and are marked read on open; a Mark all as read action clears the badge. Installed devices can enable push notifications; the service worker suppresses the system banner while the app is in the foreground. The recipient rules are described in Section 8.

5.12 Public Flight Monitor

A lobby television board at **/monitor** that requires no login. It polls the server every ten seconds and rotates through three screens every twenty seconds: On Ground, Departed, and Arrived, seven rows per page, with a page indicator. Each row shows the registration mark, trainee, instructor, destination in readable form, time of flight or departure, and elapsed or estimated time. Active NOTAMs rotate on a band every ten seconds.

Times are shown in Philippine time by default. Pressing Tab switches the screen to zulu and back, and the choice is remembered on that screen. A fullscreen control appears on mouse movement, and the F key toggles it. Colours match the dashboard board. A preview at **/monitor-preview** shows the same board filled with sample flights, for checking the layout on a TV before real data exists. Because the monitor endpoint performs a genuine database read, it doubles as the keep-alive target that stops the hosting project from being paused during breaks.

5.13 Installable App

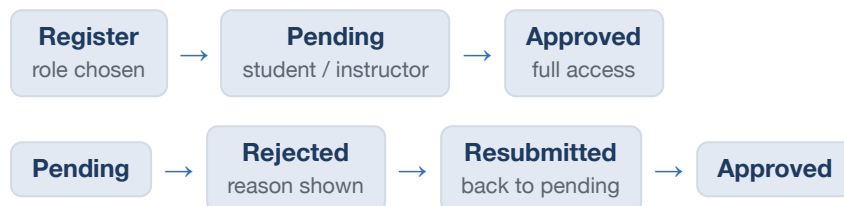
FlightraX ships a web app manifest and service worker, so it can be installed to a phone's home screen and launched standalone with the FlightraX icon and badge. Install prompts appear on the home page and in the sidebar. The service worker caches nothing on purpose (the app is auth-gated and data-driven) and exists to receive push messages and satisfy installability.

6 Business Lifecycle

This section follows every object that changes state through the system, and then puts them together as one operating day.

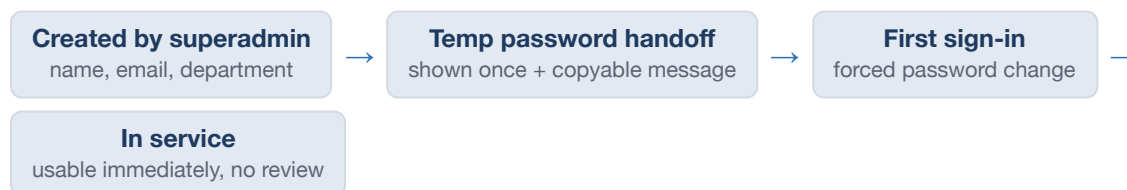
6.1 Account lifecycle

Students and instructors



1. **Registration.** A profile is created at signup, together with an account request holding the ID number and ID image.
2. **Pending.** The person is confined to the pending-approval page. Flight Operations Personnel are notified of every new registration.
3. **Review.** Flight Operations Personnel approve, or reject with a reason. The decision reaches the person by push notification (approved leads to the dashboard; rejected leads to the pending page).
4. **Resubmission.** A rejected person corrects their name, ID number, or ID image and resubmits; the request returns to pending and reviewers are told it is a resubmission.
5. **Approved.** The account gains its role's pages. A student or instructor is then nudged to add a license, a signature, and a passcode, the three things every flight needs.

Admins



An admin who forgets an unused temporary password is recovered by the superadmin regenerating it, which restarts the handoff. Once the admin has set their own password, the ordinary email reset applies like any other account.

Superadmin

The single superadmin account is created by a seeding command run against the database, never through the application. Run on a fresh database it creates the account with a documented default password and the must-change flag armed; run again later it resets the account back to that state, which is the recovery path if the superadmin credentials are ever lost. Either way, the next sign-in is met by the forced password change before anything else opens.

6.2 Fleet readiness lifecycle

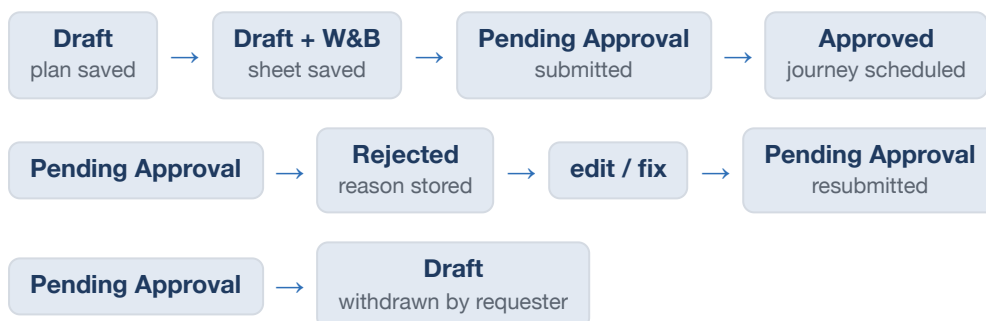


An aircraft is only offered to filers once all four steps are complete and its status is Active. Afterwards its status governs the flight lifecycle: Maintenance and Grounded block submission and approval until the aircraft is Active again; Retired asks the filer to file on another aircraft. Adding a type also adds a matching type rating that people can put on their licenses.

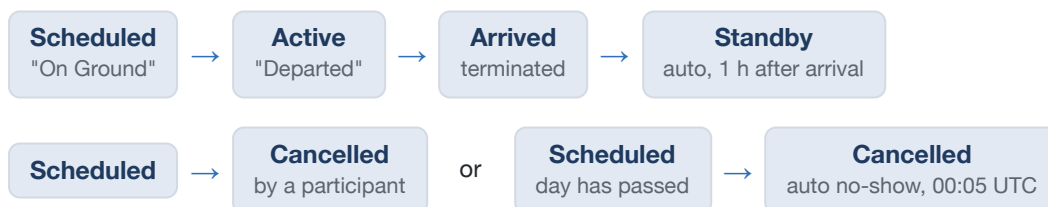
6.3 Flight lifecycle (end to end)

A flight has two linked state machines: the **request** (paperwork and permission) and the **journey** (the physical flight, created at approval).

Request states



Journey states



Step by step

1. Prepare (student or instructor)

1. Add at least one active, non-expired license, draw a signature, and set a passcode in My Account.
2. Open Flight Documents and choose an available aircraft from the picker (search by mark, filter by type).
3. Fill the flight plan, including the date of flight. Choose the flight instructor (always required) and the pilot in command: an instructor, or yourself with any valid license. The chosen people must not be marked unavailable on the date of flight.
4. Save. The plan is now a Draft with a plan code, and the Weight and Balance page opens. Endurance may still be blank at this point; it becomes mandatory at submission.
5. Fill the weight and balance sheet. The server computes totals and the weight status; you record the balance status. The sheet is signed with your signature.

2. Submit (requester)

1. Submit for approval. Refused if the sheet is missing, the endurance is blank, or the aircraft is not Active.
2. The request becomes Pending Approval. The assigned instructor and the pilot in command are each notified of a new request naming the plan code, aircraft, and zulu time.
3. While pending, the request cannot be edited. The requester may withdraw it, which returns it to Draft and tells the reviewers.

3. Review (assigned instructor or eligible pilot in command)

1. Open the request from the queue (it appears in real time) or from the notification.
2. **Approve**: enter passcode. The system checks the approver's signature and license, their eligibility to act, the aircraft's status, and that no other scheduled or active flight already holds that aircraft at the exact same date and time. It then records the approver's name, signature, and licenses on the plan and the sheet as the approving authority, creates a Scheduled journey, and marks the request Approved. The requester, the other reviewer, and the Air Traffic Controller department are notified.
3. **Reject**: enter passcode and a reason. The request becomes Rejected. The requester and the other reviewer are notified. The requester may edit and resubmit, and reviewers are told it is a resubmission.
4. A request decided by one reviewer cannot be decided again by the other.

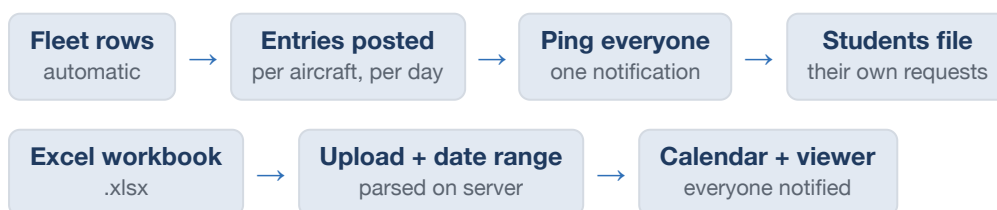
4. Fly (a participant: filer, pilot in command, or assigned instructor)

1. On the date of flight, the flight appears On Ground on the dashboard board, the public monitor, and each participant's today's flights drawer.
2. **Commence** with passcode. Refused if the aircraft is already on an active flight, if the requester, PIC, or instructor is on another active flight, or if an earlier scheduled flight on the same aircraft today has not yet commenced or been cancelled (the earlier flight is offered for cancellation). The journey becomes Active and participants are told the aircraft has departed.
3. **Terminate** with passcode when the aircraft is back. The journey becomes Arrived; participants are told. One hour later it moves to Standby automatically, clearing the aircraft from the board.
4. **Cancel** with passcode while still on ground. Participants are told who cancelled and which flight.
5. If a scheduled flight is never commenced by the end of its operations day, the nightly sweep cancels it as a no-show and tells the participants in passive voice.

5. Record

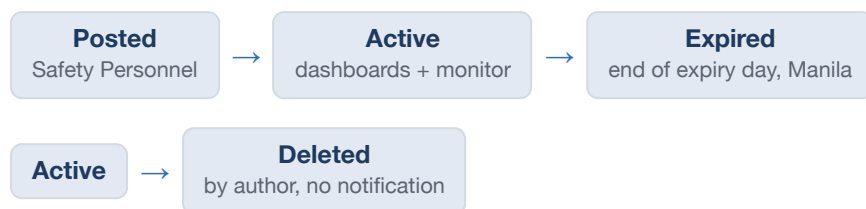
1. Arrived, standby, and cancelled flights appear in the requester's flight log and in the Air Traffic Controller's audit.
2. The flight plan and weight and balance sheet can be downloaded as PDFs at any point after approval, with the pilot in command's signature in the pilot lane.

6.4 Schedule lifecycle



The schedule is deliberately independent of flight requests. The board is posted; each student reads it and files their own request. A mismatch is on the filing, not the posting. Entries can be edited and removed at any time without notifying anyone; only the ping and the file upload announce.

6.5 NOTAM lifecycle



6.6 A day in operation

When	Who	What happens
Evening before	Flight Operations Personnel	Posts tomorrow's board (or uploads the week's workbook) and pings everyone.
Evening before	Students, instructors	Read the schedule, file flight plans against their slots, fill weight and balance, submit.
Evening / morning	Instructors	Approve or reject from the queue or from a push notification. Tower is notified of each approval.
Morning	Safety Personnel	Posts any NOTAM for the day; it appears on every dashboard and the lobby monitor.
Flight time	A flight participant	Commences the flight from the drawer; the aircraft moves to Departed on the boards.
After landing	A flight participant	Terminates the flight; it shows Arrived for an hour, then drops to standby automatically.
All day	Air Traffic Controller	Watches the organised board; audits completed flights and their documents.
All day	Anyone in the lobby	Sees the public monitor cycling On Ground, Departed, Arrived.
00:05 UTC nightly	System	Cancels any still-scheduled flight from a past operations day as a no-show and notifies participants.
Every 30 minutes	System	Moves flights that arrived more than an hour ago to standby.

7 Business Rules and Safeguards

These rules are enforced on the server, in the database, or both. They are the reason a filed document can be trusted.

7.1 Identity and authority

Rule	Detail
Signature required	Filing a plan or a sheet, and approving a request, require a stored signature. The signature is copied onto the document at that moment; later changes to the profile never rewrite it.
Valid license required	Filing and approving require at least one license that is Active and either has no expiry or expires today or later. Expired licenses count for nothing.
Passcode confirmation	Approve, reject, commence, terminate, and cancel each require the actor's four-digit passcode. Passcodes are stored as salted hashes and compared in constant time. Students and instructors hold passcodes.
Pilot in command eligibility	Anyone holding any valid license may be named pilot in command, a student on a Student Pilot License included. Being PIC does not by itself grant approval rights: an instructor PIC may decide with any valid license, while a student PIC may self-approve only with a valid Private Pilot License. An SPL-only student flies under an approval from the assigned instructor.
Who may decide a request	The assigned flight instructor always may. The pilot in command may only when eligible as above. Nobody else may, including instructors browsing the "All" queue.
Instructor always required	Every plan names a flight instructor, who must hold the instructor role. Choosing yourself as PIC does not remove this requirement.
Ownership	Only the requester can edit, delete, submit, or withdraw their plan. Only a flight's participants — its filer, pilot in command, or assigned instructor — can commence, terminate, or cancel it; the today's flights drawer shows each person only those flights. The superadmin can see and act on all.
Administrative provisioning	Admin accounts exist only through the superadmin, and the superadmin only through the seeder. Temporary passwords are random, shown once, and force a change on first sign-in; nothing privileged ever ships with a password chosen by a form.
Role-locked login	Credentials are accepted only on the login page for the account's role, and logout returns to that same page.
Unique identity	A student or instructor ID number can belong to only one account, and an email address can register only once.

7.2 Document state rules

- Only Draft or Rejected plans may be edited, deleted, or submitted. Deleting a plan removes its request and sheet.

- Submission requires the weight and balance sheet to exist and the endurance to be filled; a draft may be saved without either. The sheet can be filed only while the request is Draft or Rejected.
- Only Pending Approval requests may be approved, rejected, or withdrawn. A withdrawn request returns to Draft rather than being deleted.
- Approval and resubmission clear any earlier rejection reason and rejecting reviewer.
- Plan codes are unique; a rare same-day collision is retried automatically.

7.3 Aircraft and scheduling rules

Rule	Detail
Aircraft readiness	Only Active aircraft with complete type specifications and a basic empty weight configuration are offered to filers. Drafts on an aircraft are never blocked by an active flight; the guards live at submit and approval.
Aircraft status blocks	Submission and approval refuse Maintenance and Grounded aircraft ("wait until it is active again") and Retired aircraft ("file on another aircraft"). A deleted aircraft asks the filer to delete the plan.
One flight per aircraft per DOF	An aircraft cannot hold two scheduled or active journeys with the identical date and time of flight. Approval refuses with the other filer's name; the database enforces it as well, so two simultaneous approvals cannot both succeed.
One aircraft airborne at a time	A flight cannot commence while that aircraft is on another active flight. Enforced by the database too.
One flight per person at a time	A flight cannot commence while its requester, pilot in command, or instructor is on another active flight.
Flights commence in order	An earlier scheduled flight on the same aircraft today must commence or be cancelled before a later one starts. The later flight's requester is offered the earlier flight for cancellation if they may cancel it.
Instructor availability	A person marked unavailable on the date of flight cannot be chosen as PIC or instructor. Unavailability periods cannot overlap or lie wholly in the past.
Schedule overlap	Two schedule entries on one aircraft cannot overlap in time; back-to-back is allowed. Two uploaded files cannot cover the same date.

7.4 Time conventions

- Operations day is Philippine time; clock times are zulu.
- The date and time of filing is DDHHMM, where DD is the local (Asia/Manila) calendar date and HHMM is the zulu clock time. The departure time field is pre-filled to one hour after the filing time.
- The date of flight is its own field in ICAO YYMMDD form. It prints as the DOF/ line in Other Information and is the day that instructor availability, aircraft conflicts, the boards, the today's flights drawer, and the no-show sweep all key on. "Today" everywhere means the Manila date.
- NOTAM expiry is the end of the day in Manila. Departure times, EET, and endurance are zulu and are always shown with a Z suffix. The public monitor shows Philippine time by default and can be switched to zulu.

7.5 Audit and snapshots

- Filed forms are audit records. The aircraft's identification, type, designator, and markings; the pilot's name, signature, and licenses; and the approver's name, signature, and licenses are copied onto the plan at filing and approval. The sheet snapshots the aircraft's and type's figures and the arms in force at the time.
- The printed plan's pilot lane carries the pilot in command from the filing snapshot only once the request is approved — no signature unless approved; unapproved plans print unsigned under a DRAFT watermark. The duly authorised representative lane is never printed. Who approved, and with what signature and licenses, is snapshotted on the record itself, so the authorisation survives in the database even though the client's form convention keeps it off the printout.
- Every request records who approved or rejected it and when. Every journey records who commenced, terminated, or cancelled it and when. A cancellation with no actor is, by definition, a system no-show.
- Account reviews record the reviewer, the timestamp, and the reason.

7.6 Plain-language errors

Every message a user sees is a sentence describing what to do next. Validation messages are written by hand for each field. Database and provider errors are translated to plain English on the server and the raw detail is logged, never shown. Uploads that are too large are shrunk in the browser before they leave the device.

7.7 Repeated and duplicate submissions

On a slow connection people tap a button twice, or retry after the first response never arrives. FlightraX treats the repeat as the same submission at two levels.

Level	Behaviour
On the device	While a request is running, further presses are ignored and the button stays disabled for the whole request. Forms that leave the page once they succeed (sign-in, registration, password reset, approve, reject, submit, create or delete a flight plan) stay locked until the next page has loaded.
Decisions and flight actions	A repeated approve, reject, submit, withdraw, commence, terminate, or cancel by the same person reports success instead of an error. A status change only applies to a record still in the expected status.
New records	An identical flight plan, NOTAM, certificate, schedule file, or schedule entry from the same person within five minutes is treated as the same submission. No second copy is made and nobody is notified twice.
Schedule ping	A second ping of the same date by the same person within one minute is ignored, so the whole school is not notified twice.
Natural duplicates	A registered ID number or email, an existing license type and number, or an identical unavailability period is recognised and answered with a clear message.

8 Notifications and Communication

Notifications are how FlightraX replaces "someone walks over and tells you". They are composed inside the database at the moment a row changes, which means every path that changes a flight, including the overnight sweep, produces the same message.

8.1 Four rules that govern every notification

Rule	Meaning
A. Never the actor	A person is never notified about something they did themselves.
B. Composed per recipient	Each row stores its finished title and body. When a person acted, they are named ("Capt. Reyes cancelled your flight FP-3K9Q2 (RP-C1234, 0800Z)"). When the system acted, the passive voice is used ("Your flight ... was automatically cancelled (no-show)"). A flight message always identifies the flight by plan code, registration, and zulu time.
C. Per-flight events never broadcast	Commenced, arrived, cancelled, no-show, and aircraft status changes reach only that flight's participants: trainee, pilot in command, and assigned instructor. Fleet-wide awareness is the monitor's job.
D. One row per person per event	Recipient sets are deduplicated. A person who is trainee, PIC, and instructor on one flight receives one notification.

8.2 Event matrix

Event	Triggered by	Recipients
Account submitted	Student or instructor registration	Flight Operations Personnel
Account resubmitted	Rejected account resubmits	Flight Operations Personnel
Account approved	Account review	The person (push; opens the dashboard). Any earlier rejection notice is marked read.
Account rejected	Account review	The person (push; opens the pending page with the reason)
Flight request submitted	Submit (or resubmit, which is labelled as such)	Assigned instructor and pilot in command, each told which role they hold
Flight request approved	Approve	Requester, the reviewer who did not act, and the Air Traffic Controller department
Flight request rejected	Reject	Requester and the reviewer who did not act, with the reason
Flight request withdrawn	Requester cancels a pending request	Assigned instructor and pilot in command
Flight commenced	Commence	Participants

Flight arrived	Terminate	Participants
Flight cancelled	Cancel	Participants
Flight no-show	Nightly sweep	Participants (passive voice)
NOTAM posted	Safety Personnel	Everyone except Flight Operations Personnel
Aircraft status changed	Fleet status change	Participants of each live flight on that aircraft, one message per flight
Instructor unavailable	Unavailability added	All students
Schedule ready	Ping everyone	Everyone (opens that date's board)
Schedule file uploaded	Workbook upload	Everyone (opens the workbook viewer)

No notification is sent for private drafts, sheet saves, credential edits, profile changes, password or passcode changes, admin account creation and password regeneration (those hand over credentials in person), fleet housekeeping, schedule entry edits, NOTAM deletion, unavailability removal, or the automatic standby transition.

8.3 Delivery channels

- **In-app.** The bell badge and panel update in real time through a per-user subscription, so each event is delivered to exactly one client. A full page lists history with mark-read controls.
- **Push.** Every new notification row is handed asynchronously to the push sender, which delivers to every device the person has enabled, using standard Web Push with VAPID keys. Dead subscriptions are pruned automatically. The banner is suppressed while the app is in the foreground, up to a safety limit. Pending accounts may enable push so they learn their review outcome.
- **Email.** Used only for password reset (and signup confirmation if enabled), through a verified sending domain.

9 Technical Architecture

9.1 Stack

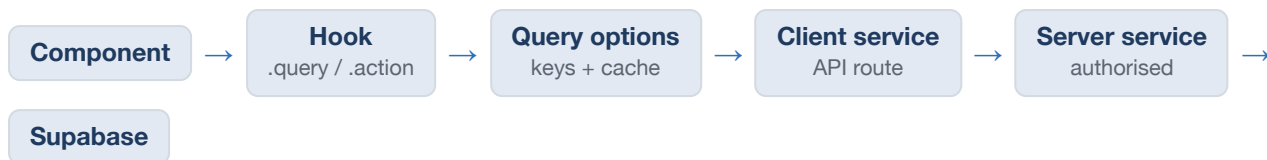
Layer	Technology
Framework	Next.js 16 (App Router) on React 19, TypeScript 5 in strict mode
Styling and UI	Tailwind CSS 4, shadcn/ui on Radix primitives, Lucide icons, Motion for animation
Backend	Supabase: Postgres, Auth, Storage, Realtime, pg_cron, pg_net
Mutations	next-safe-action server actions with Zod 4 validation, behind a shared single-flight guard
Server state	TanStack Query 5 (client cache), TanStack Table 8 (server-paginated tables)
Forms	React Hook Form with Zod resolvers
UI and URL state	Zustand 5 (UI only), nuqs (URL-driven filters, tabs, pagination)
Documents	pdf-lib (flight plan and W&B PDFs in the browser), pdf.js legacy build (in-app PDF preview on every device), exceljs (schedule workbook parsing on the server), signature_pad (signature canvas)
Notifications	Supabase Realtime, Web Push via web-push, service worker
Hosting	Vercel (application), Supabase hosted project (database and services), Resend (transactional email via SMTP)
Domain	flightrax.app (user-facing); flightrax.vercel.app (machine address for keep-alive and push dispatch)

9.2 Structure

The codebase separates routing from logic and keeps every feature in its own module:

- **app/** contains only routes and thin API handlers; each page renders a module's root component.
- **modules/** holds one folder per feature (auth, account-review, admins, aircrafts, dashboard, flight-documents, instructors, students, schedule, notams, notifications, monitor), each with components, actions, hooks, queries, services, schemas, types, and utils. Modules never import from one another.
- **shared/** holds cross-cutting code: the role and permission model, route guards, Supabase clients, storage bucket names, aviation helpers (license validity, ratings, board formatting, time zone), the submission guard, layout shell, and generated database types.
- **supabase/migrations/** holds the entire database history as SQL, including functions, triggers, policies, cron jobs, and realtime publications.

9.3 Data flow



Reads go through TanStack Query so every surface shares one cache. Writes go through server actions, each of which validates input with a Zod schema, loads the acting profile, checks approval and permission, performs the change with the service-role client, and returns a plain-language result. Every mutation hook runs through a shared guard that allows one request at a time. After a successful mutation the hook invalidates the affected query keys so the UI refreshes reactively. Tables paginate on the server with page state in the URL.

9.4 Security model

- **Session and routing.** A Next.js proxy refreshes the Supabase session on every request, redirects signed-out users to login, confines unapproved accounts to the pending page, and redirects anyone lacking a route's permission to their dashboard. The same permission table drives navigation. The superadmin's reach is additionally shaped by the `superadmin_full_navigation` setting, enforced in the same proxy and in the sidebar, so confining the account is a data change, not a deploy.
- **Authorisation in actions.** Every action and server service re-derives the actor from the session and checks approval, role, and department on the server. Client-side checks are only for hiding controls. Superadmin-only actions (creating an admin, regenerating a temporary password) are guarded by a permission no role holds.
- **Administrative credentials.** Temporary passwords are generated server-side from an unambiguous alphabet, returned exactly once to the superadmin, and never stored in plain text anywhere. The must-change flag rides on the profile and is cleared only by the owner setting a new password; the forced-change dialog cannot be dismissed, clicked around, or escaped.
- **Database access.** Client code never talks to Supabase tables directly. Server code uses the service-role client for writes; row-level security policies still shape what authenticated clients may read (own rows, staff views of students, and so on) and what realtime delivers to each subscriber. The `app_settings` table is readable by any authenticated user and writable by no one but server code.
- **Secrets.** The service-role key, VAPID private key, and push dispatch secret live only in server environment variables. The push dispatch endpoint requires a shared secret compared in constant time.
- **Passcodes** are scrypt-hashed with a per-user salt. Signatures are stored as SVG text.
- **Storage.** Account ID documents are private and served by short-lived signed URLs to the owner and reviewers. Profile and aircraft photos are public. License and certificate images are readable by the owner and by staff.
- **Public surface.** The monitor endpoint exposes only display fields (registration, names, destination, times, NOTAM text) and no identifiers that lead anywhere.

9.5 Database-side logic

- **Functions** compute the dashboard board, the participant-scoped today's flights list, the public monitor board, and the flight plan aircraft picker in single queries with pagination and search.
- **Triggers** maintain updated-at stamps, derive each journey's operations date from its date of flight, create profile rows at signup, compose and fan out notifications, and hand each new notification to the push sender through `pg_net`.

- **Constraints** enforce ICAO field formats, status consistency, unique registration marks, plan codes, and ID numbers, one flight per aircraft per date and time, one active flight per aircraft, and non-overlapping schedule entries, unavailability periods, and uploaded files.
- **Scheduled jobs** (pg_cron): every 30 minutes, arrived flights older than one hour move to standby; nightly at 00:05 UTC, scheduled flights from a past operations day are cancelled as no-shows.
- **Realtime** is published for flight requests, flight journeys, aircraft availability, NOTAMs, and notifications. A shared client hook attaches the user's token, subscribes once per table per page, invalidates the relevant queries on events, and falls back to 15-second polling if the socket cannot connect.

9.6 Documents and files

- Flight plan PDFs are produced by overlaying text and signature vectors onto the official CAAP Form ATS 2019-1 shipped with the application; the weight and balance sheet is drawn from scratch. Both are built in the browser from a single export payload.
- The in-app preview renders every page to a canvas with the PDF renderer's legacy build, plus the one modern-JavaScript polyfill that build still needs, so the same preview works on older phone and tablet browsers that the standard build refuses. There is no embedded-viewer fallback: one rendering path on every device.
- Schedule workbooks are parsed on the server into JSON grids (values, merges, fills, alignment, column widths) and stored per sheet, with the original kept in storage for download.
- Images are compressed client-side before upload to stay under the hosting request limit.

10 Data Model

The schema is small and deliberately denormalised where documents need to be audit records. Tables are listed by domain.

Table	Purpose
Identity	
profiles	One row per user: name, email, role, photo, signature, passcode hash, and the must-change-password flag that arms the forced first-login change.
student_profiles / instructor_profiles / admin_profiles	Role-specific one-to-one rows; admin_profiles carries the department. The superadmin needs no side table: its role alone carries its authority.
account_requests	Verification request for students and instructors: ID number, ID document, status, reviewer, reason, submission time.
licenses	A user's licenses: type, number, ratings, expiry, status, front and back images.
certificates	A user's certificates: title, description, expiry, image.
instructor_unavailabilities	Leave periods per instructor, non-overlapping.
push_subscriptions	Web Push endpoints per user device.
Fleet	
aircraft_types	Type catalogue with ICAO designator and W&B specifications.
aircraft_type_baggage_areas	Up to six baggage positions per type with arms.
aircrafts	Fleet: registration mark and number, type, serial, markings, photo, status.
aircraft_weight_balance_configs	Basic empty weight, arm, and moment per aircraft.
Flight documents and operations	
flight_plans	The CAAP flight plan with all ICAO fields, the filing time and the date of flight with their resolved timestamps, plan code, and snapshots of aircraft, pilot, licenses, signatures, and the approving authority.
weight_balances	The W&B sheet snapshot: givens, inputs, computed totals, statuses, preparer and verifier.
weight_balance_baggage_entries	Per-position baggage weight, arm, and moment for a sheet.
flight_requests	Bundles plan and sheet; status, requester, assigned instructor, approver or rejecter, reason.
flight_journeys	The physical flight after approval: status, aircraft, date and time of flight, operations date, commenced/terminated/cancelled by and at.
Schedule and notices	

schedule_entries	A block on the board: aircraft, time range, session type, pilot, instructor, or note.
schedule_pings	Each "ping everyone" with date, sender, and time.
schedule_uploads	An uploaded workbook: file, storage path, date range, uploader.
schedule_upload_sheets	Each sheet of an upload as a parsed grid with its board date.
notams	Notices: title, description, severity, expiry, author.
notifications	One row per recipient per event with composed title, body, link, and read state.
Configuration	
app_settings	Global settings as one row per key with a JSON value; currently superadmin_full_navigation. Authenticated users may read; only server code writes.

Enumerations

Roles	student, instructor, admin, superadmin
admin_department	flight_operations_personnel, air_traffic_controller, safety_personnel
approval_status	pending, approved, rejected
aircraft_status	active, maintenance, grounded, retired
license_status	active, expired
request status	draft, pending_approval, approved, rejected
journey_status	scheduled, active, arrived, standby, cancelled
weight / balance status	within_limits, overweight / balanced, nose_heavy, tail_heavy

Storage buckets

account-documents	Private. ID images for account review (5 MB, JPG/PNG/WebP).
profile-photos	Public. Avatars.
aircraft-photos	Public. Fleet photos.
license-images	Front and back license images; owner and staff read.
certificate-images	Certificate images; owner and staff read.
schedule-files	Original uploaded workbooks for download.

11 Operating Environment and Constraints

FlightraX is designed to run at zero cost on the free tiers of its providers. The limits below shape the design and are the signals for when an upgrade would be needed.

11.1 Hosting limits that matter

Limit	Allowance	Design response
Realtime concurrent connections	200	One websocket per open tab, multiplexed across channels. The public monitor polls instead of subscribing. This is the binding limit: plan an upgrade if simultaneous open sessions approach 150.
Realtime messages per month	2,000,000	Per-user filters on notifications, one subscription per table per page, events carry no data (they trigger refetches). Not a practical risk.
Database size / file storage	500 MB / 1 GB	Images compressed client-side; workbooks capped at 4 MB; parsed grids stored compactly.
Inactivity pause	After 7 quiet days	A scheduled job pings the monitor endpoint twice a week so the project never idles through a semester break.
Automatic backups	None on the free tier	Manual dump routine documented; backups to be taken before breaks and destructive changes and kept outside the repository.
Transactional email	3,000 / month, 100 / day, verified domain required	Only password reset uses email. Sending is from the verified flightrax.app domain.
Application hosting (Hobby plan)	Non-commercial use; no organisation repositories	Appropriate for a capstone. Adoption for real operations, or deployment from the organisation account, requires the paid application hosting tier.

Indicative cost of removing every limit above: roughly USD 45 per month across the two providers.

11.2 Configuration outside the code

- Supabase authentication: site URL and redirect allowlist for flightrax.app, flightrax.vercel.app, and local development; custom email templates carrying a token hash; custom SMTP through Resend.
- Push: VAPID key pair, subject, and dispatch secret in the application environment; a single configuration row in the database pointing the dispatch trigger at the application's endpoint.
- Application settings: the app_settings table in the database, currently the superadmin_full_navigation flag (shipped off). Flipping the row changes the superadmin's navigation and routes on the next load, with no deploy.
- Superadmin seeding: a documented command (npm run seed:superadmin) run by the operator creates the superadmin on a fresh database or resets it as a recovery path; the email and initial password can be

overridden through environment variables, and the account always demands a new password at the next sign-in.

- Keep-alive: a repository variable holding the application URL.

11.3 Devices and browsers

- Desktop browsers and Android Chrome are the verified targets, including installed-app mode and push.
- iOS is supported in code (install-first gate, custom permission prompt, Apple push) but unverified on a device.
- PDF previews use a compatibility rendering path verified against the older JavaScript engines found in installed-app webviews on phones and tablets.
- The public monitor is designed for a television browser: large type, fixed layout, no scrolling, fullscreen.

12 Current Status and Roadmap

As of 27 September 2026 every module described in this document is built and deployed to the hosted environment. There is no production data yet; test accounts exist for each role. The project is in end-to-end QA across every role, page, and rule.

12.1 Delivery timeline

Period	Milestone
Late June 2026	Project set up. Roles, profiles, and account review foundation.
July 2026	Aircraft management, profile photos, signatures and passcodes, NOTAMs.
Early August 2026	Licenses and certificates, centralised account requests for students and instructors.
Mid-to-late August 2026	Weight and balance specifications, flight plans, flight requests, flight journeys, dashboard boards, plan codes, instructor unavailability, standby and no-show automation.
Early September 2026	Operations-day time model, ICAO designators, public flight monitor, mandatory instructor and PIC, in-app notifications, push notifications, password reset.
12 to 14 September 2026	Schedule board, ping everyone, Excel schedule uploads with calendar and viewer.
15 to 17 September 2026	Protection against double taps and retries on every form, plain-language errors on every submission path, and Philippine time as the monitor default.
22 to 27 September 2026	Dedicated date of flight driving all scheduling, the printed pilot lane fixed to the pilot in command, participant-scoped flight controls, PDF preview compatibility for older devices, and the superadmin role: seeded account, admin provisioning with one-time temporary passwords and a forced first-login change, the Admins directory, the navigation setting, and removal of all administrative self-registration.

12.2 Open items and next steps

Item	Status
End-to-end QA pass on phone and desktop	In progress.
iOS push and install verification	Pending a device.
Schedule uploads browser pass	Built; visual verification pending.
Manual backup routine	Documented; to be scheduled before the first break.
User manual and training guide	Next document after this one.
Hosting tier decision	Hand-over condition: paid application hosting if WCC adopts FlightraX for real operations.

12.3 Ideas considered for later

- Additional schedule views and richer row editing on the board.
- An audit feed of fleet housekeeping changes for Flight Operations Personnel.
- Further global settings behind the app_settings table as operating policy firms up.
- A single canonical domain once push dispatch and keep-alive are repointed.

13 Glossary

Term	Meaning in FlightraX
CAAP Form ATS 2019-1	The Philippine Civil Aviation Authority's ICAO flight plan form. FlightraX's flight plan mirrors its fields and prints onto it.
Flight plan	The filed ICAO flight plan for one flight.
Weight and balance (W&B)	The loading sheet proving the aircraft is within its takeoff weight and balanced.
Flight request	The bundle of plan plus sheet that moves through approval.
Flight journey	The physical flight created when a request is approved: scheduled, active, arrived, standby, or cancelled.
Plan code	The unique identifier of a flight plan, FP-YYMMDD-NNNN, shown on cards and in every flight notification.
Date and time of filing	Filed as DDHHMM: local day, zulu time. The departure time defaults to one hour after it.
Date of flight (DOF)	The day the flight happens, filed as YYMMDD. Prints as the DOF/ line in Other Information and drives availability, conflicts, the boards, and the no-show sweep.
Zulu (Z)	Coordinated Universal Time. All clock times on forms are zulu.
Operations day	The Philippine (Asia/Manila) calendar date used for "today".
EET	Estimated elapsed time of the flight, HHMM.
PIC	Pilot in command: anyone named on the plan holding any valid license, a Student Pilot License included. Approval authority is narrower — see Section 7.1.
Flight instructor (FI)	The instructor assigned to a request; always required; always a reviewer.
Authorised representative	The approver's snapshot (name, signature, licenses) stored on the plan at approval. Kept as a database audit record; the printed form's representative lane is never filled — the pilot lane always shows the PIC.
Participants	A flight's trainee (filer), pilot in command, and assigned instructor — the only people who can commence, terminate, or cancel it.
Passcode	A user's four-digit security code confirming flight decisions and operations.
Signature	A drawn vector signature stored on the profile and copied onto documents.
License validity	Active status and not expired (no-expiry or expiry date today or later).
Ratings	Instrument Rating plus one type rating per aircraft type in the fleet.
SPL / PPL / CPL / FI	Student, Private, Commercial Pilot License; Flight Instructor License.
Superadmin	The single seeded system-owner account that provisions admin accounts and holds every permission.

Temporary password	A random, single-handoff password issued when the superadmin creates or recovers an admin account; the first sign-in must replace it.
NOTAM	Notice to Airmen: an operational notice with severity and expiry.
ZZZZ	ICAO placeholder for an aerodrome without a code; here, the school's home field RPT-20 Binalonan.
Other Information	Item 18 of the flight plan: DOF, DEP, DEST, ALTN lines and license lines, auto-filled.
On Ground / Departed / Arrived	The three board sections, mapping to scheduled, active, and arrived journeys.
Standby	The automatic state one hour after arrival that clears the aircraft from boards.
No-show	A scheduled flight never commenced by the end of its operations day; cancelled automatically.
Ping	The one-time announcement that a day's schedule board is ready.
Session types	PS, TAXI, LCL, XC, IRS, CAAP, CF, OTS, TBD, CT, UPRT: the legend of the schedule board.
Flight Ops / FOP	Flight Operations Personnel, the operations desk admin department.
ATC	Air Traffic Controller, the tower desk admin department.
Safety	Safety Personnel, the admin department that posts NOTAMs.
PWA	Progressive web app: the installable, push-capable form of the web application.

A Appendix A. Routes and Surfaces

Public and authentication

Route	Surface
/	Home page with sign-in, register, and install prompt. The FlightraX logo doubles as the superadmin's hidden entry.
/login, /login/student, /login/instructor, /login/admin, /login/superadmin	Role picker and role-locked sign-in pages. The admin page is reached by the hold-to-fly plane; the superadmin page by tapping the home logo five times.
/register, /register/student, /register/instructor	Role picker and registration forms. Students and instructors only; administrative roles have no registration.
/forgot-password, /reset-password	Password recovery.
/pending-approval	Waiting or rejected account state with resubmission and push opt-in.
/monitor	Public lobby monitor with live data.
/monitor-preview	The same monitor filled with sample flights, for layout checks.

Dashboard

Route	Surface
/dashboard	Role-dependent home: flight board, organised board, or aircraft manager.
/account	My Account.
/flight-documents	Own flight documents, grouped by status.
/flight-documents/flight-plans/new	Aircraft picker and new flight plan.
/flight-documents/flight-plans/[id]	Edit a draft or rejected plan.
/flight-documents/flight-plans/[id]/weight-balance	Weight and balance sheet.
/flight-documents/flight-plans/[id]/log	Flight log for one flight: plan, sheet, journey, download.
/flight-requests, /flight-requests/[flightPlanId]	Instructor review queue and review page.
/flight-plans, /flight-plans/[id]	Air Traffic Controller audit and flight log.
/schedule, /schedule/uploads/[id]	Schedule board and files calendar; workbook viewer.
/instructors, /students	Directories.
/admins	Superadmin-only Admins directory: create admins, regenerate temporary passwords.
/account-review	Account verification queue.

/notams	NOTAM management.
/notifications	Notification history.

API routes (read paths and integrations)

Account review queue, metrics, and document; aircraft types, baggage areas, aircraft, and weight and balance configuration; admins directory; app settings; auth profile, licenses, certificates, images, flight logs, and email confirmation; aviation rating options; dashboard flight status, today's flights, and active NOTAMs; flight document aircraft options, filer context, PIC and type options, flight plans, journeys, weight and balance, export, requests, and review requests; approved instructors and students; monitor board; NOTAMs; notifications, unread count, and push dispatch; schedule day, people, and uploads.

B Appendix B. Server Actions Inventory

Every state change in FlightraX goes through one of these validated server actions. Each checks the actor's session, approval, and permission before writing.

Module	Actions
Auth and account	login, logout, registerStudent, registerInstructor, requestPasswordReset, resetPassword, changePassword, setInitialPassword (the forced first-login change), resubmitRejectedAccount, uploadProfilePhoto (and remove), saveSignature, savePasscode, createLicense, updateLicense, deleteLicense, createCertificate, updateCertificate, deleteCertificate
Account review	approveAccountRequest, rejectAccountRequest
Admins (superadmin)	createAdmin, regenerateAdminPassword
Aircrafts	createAircraftType, deleteAircraftType, setAircraftTypeWbSpecs, createAircraft, updateAircraft, updateAircraftStatus, deleteAircraft, setAircraftWeightBalance
Flight documents	createFlightPlan, updateFlightPlan, deleteFlightPlan, saveWeightBalance, submitFlightRequest, cancelFlightRequest (withdraw), approveFlightRequest, rejectFlightRequest
Dashboard (operations)	commenceFlight, terminateFlight, cancelFlight
Instructors	addInstructorUnavailability, removeInstructorUnavailability
Schedule	createScheduleEntry, updateScheduleEntry, deleteScheduleEntry, pingScheduleReady, uploadScheduleFile, deleteScheduleUpload
NOTAMs	createNotam, deleteNotam
Notifications	markNotificationRead, markAllNotificationsRead, savePushSubscription, deletePushSubscription
Shared	verifyPasscode

The former registerAdmin and superadmin registration actions were removed with the registration pages in v1.3; provisioning replaced them.

Scheduled jobs and commands

Job	Schedule	Effect
standby-arrived-flights	Every 30 minutes	Arrived journeys terminated more than one hour ago move to standby.
cancel-no-show-flights	Daily 00:05 UTC	Scheduled journeys whose operations day has passed are cancelled with no actor; participants are notified.
Keep Supabase Awake (GitHub Actions)	Monday and Thursday 03:00 UTC	Requests the public monitor endpoint to keep the hosted project active.

seed:superadmin (operator command)	On demand	Creates the superadmin on a fresh database, or resets it as a recovery path; the next sign-in must set a new password.
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C Appendix C. Reference Data

Flight plan field formats

Field	Accepted format
Date and time of filing	DDHHMM; day 01 to 31, time in zulu.
Date of flight	YYMMDD (ICAO Item 18 DOF/).
Flight rules	I (IFR), V (VFR), Y (IFR then VFR), Z (VFR then IFR).
Type of flight	S scheduled, N non-scheduled, G general aviation, M military, X other.
Number of aircraft	1 to 999.
Wake turbulence category	L light, M medium, H heavy.
COM/NAV equipment	N none, S standard.
Surveillance equipment	N none, A transponder Mode A, C transponder Mode A and C.
Aerodromes	Four-letter ICAO code from the Philippine list, or ZZZZ. Alternates optional.
Departure time	HHMM zulu; defaults to one hour after the filing time.
Cruising speed	K or N plus four digits, or M plus three digits (for example N0110).
Cruising level	VFR, or F/A plus three digits, or S/M plus four digits (for example A045).
Total EET, endurance	HHMM. Endurance is required at submission, optional on a draft.
Persons on board	Three digits greater than 000, or TBN.
Emergency and survival equipment	Marked paper-style: cross out what is not carried; unmarked means carried.
Dinghies	Number up to 99, capacity up to 999, covered flag, colour.

Weight and balance computation

- Total weight = basic empty weight + usable fuel weight + instructor and student weight + sum of baggage weights
- Total moment = basic empty weight moment + usable fuel moment + instructor and student moment + sum of baggage moments
- Centre of gravity = total moment $\div$ total weight
- Weight status = Within Limits when total weight $\leq$ the type's maximum takeoff weight, otherwise Overweight
- Balance status = recorded by the preparer: Balanced, Nose Heavy, or Tail Heavy

Schedule session types

Code	Label	People or note
PS	Pre-Solo Simulator	People

TAXI	Taxi Exercise	People
LCL	Local Flight	People
XC	Cross Country Flight	People
IRS	Instrument Simulator	People
CAAP	CAAP Checkride	People
CF	Company Flight	People
OTS	Out of Service	Note
TBD	To Be Determined	Note
CT	Cross Trainees	People
UPRT	Upset Prevention and Recovery Training	People

Upload limits

ID, license, certificate, profile, aircraft images	JPG, PNG, or WebP; 5 MB maximum; shrunk in the browser above 1 MB.
Schedule workbook	.xlsx only; 4 MB maximum; up to two months per file; up to 300 rows and 80 columns per sheet.
NOTAM text	Title 200 characters; description 300 characters.
Rejection reasons	Account: at least 3 characters. Flight request: 1 to 1000 characters.

Temporary password format

Generated as tmp- followed by two four-character groups drawn from an alphabet with no ambiguous characters (no 0/o, 1/l/i), because these passwords get read aloud and retyped once and then thrown away.

FLIGHTRAX: Project Application, v1.3, 27 September 2026. Prepared by the FlightraX Team (Researchers): MANZANO, KYLE CEDRICK G.; GAÑALONGO, CHRISTAN BENEDICT G.; JOVES, DANIELA GEMAICA Z.; OSAYAN, GIAN CARLO V.; REYES, JUSTINE SHIN R.; ABDON, DENMARK S. Based on the application source, database migrations, and project documents at application version 0.1.0. Supersedes v1.2 (22 September 2026); this revision adds the superadmin role and administrative account provisioning, and updates the flight document and operations rules to match.